PRODUCT CONCEPT & SCOPE

Protein & Supplement Retail SaaS Platform

Multi-store business management, point-of-sale billing, inventory and profit tracking, plus a customer loyalty app for independent retailers.

Business platform Owner dashboard, branch management, inventory, suppliers and analytics	**POS & billing** Fast checkout, rupee/percentage discounts, receipts, printing and WhatsApp sharing
Customer loyalty Join with a shop code or QR, discover offers, earn points and redeem rewards	**SaaS administration** Retailer onboarding, subscription plans, access control and platform reporting

Product goal: help supplement retailers operate their stores efficiently, protect financial data, understand real product profitability, and build repeat customer relationships.

PRODUCT DESCRIPTION • VERSION 1.0 • 4 OCTOBER 2026

1. Product Overview

The product is a multi-tenant SaaS platform for protein powder, sports nutrition and supplement retailers. Each independent business can create multiple shop branches, invite multiple admins and billing staff, manage inventory and sales, and view branch-level or consolidated analytics. Customers can join a shop's loyalty programme through a unique shop code or QR code and receive shop-specific offers and rewards.

Core applications

• **Super Admin Portal:** manage retailer accounts, subscription plans, account status, platform usage and support.

• **Owner/Admin Web App:** manage branches, products, purchase costs, selling prices, stock, team permissions, offers and reports.

• **Point of Sale (POS):** product lookup/barcode scanning, cart, discount entry, checkout, invoice generation, printing and WhatsApp sharing.

• **Owner Mobile App:** monitor sales, stock alerts and branch performance; additional mobile billing can be added to scope.

• **Customer Mobile App:** join participating stores, browse offers, view rewards and purchase history, and access membership QR codes.

2. Roles & Permissions

Access is role-based and must be enforced by the backend API—not merely by hiding buttons in the user interface.

Capability	Owner	Admin	Shop Boy / Cashier
Branches	All business branches	Assigned branches	Assigned shop only
Products & prices	Full control	If permitted	View product and selling price
Purchase cost	View/edit	Permission-based	Never visible
Profit & financial reports	All branches	Permission-based	Not visible
Create bills	Yes	Yes	Yes
Team & settings	Full control	Permission-based	No
Discounts	Configure policy	Approve/manage if permitted	Apply within configured limits

Owner can create multiple admins and cashiers, assign them to selected branches, configure granular permissions, disable access, and review an audit log of price changes, stock adjustments, discounts, refunds and cancellations.

3. Inventory & Purchase Cost Management

• Product catalogue with name, brand, category, SKU/barcode, variant/size, image and tax configuration.

• Purchase records with supplier, invoice reference, quantity, unit cost, batch/lot, expiry date and other allocated costs where needed.

• Selling price management with effective price changes and a record of who made each change.

• Stock ledger recording purchases, sales, returns, damage, adjustments and transfers between branches.

• Low-stock and near-expiry alerts; branch-specific stock availability.

• Inventory valuation and profit reporting using a clearly defined costing method, such as weighted-average cost or FIFO.

Important: new purchases must not overwrite historical costs. The system should retain purchase history and calculate cost of goods sold consistently so profit reports remain explainable.

4. POS, Billing & Discounts

• Search by product name, SKU or barcode; add multiple products and quantities to a cart.

• Apply a discount either as a fixed currency amount (■) or as a percentage (%). Support item-level discounts and bill-level discounts.

• Owner-configured cashier limits; require admin approval for discounts above the permitted threshold.

• Validate discounts on the backend; prevent negative totals and discounts beyond the eligible amount.

• Record subtotal, discount type/value, tax calculation, final total, payment method, cashier and approval details.

• Generate invoice/receipt PDF, print on compatible 58 mm or 80 mm thermal printers, and share the bill through WhatsApp.

• Support invoice cancellation/refund workflows with permission checks and audit history.

Tax behaviour and invoice format must be configured to meet the retailer's applicable requirements. GST invoices and reports are included in the planned scope, subject to final tax rules and implementation validation.

5. Customer App & Loyalty Programme

• Each shop or branch can have a unique join code, QR code and signup link.

• A customer can use one account to join multiple participating stores, with each store's membership and benefits kept separate.

• Customer profile, membership QR code, participating shops, purchase history and available rewards.

• Shop-specific offers, coupon codes, welcome promotions and loyalty points.

• Configurable earn/redeem rules, minimum spend, eligibility, expiry, usage limits and exclusions.

• POS can associate a sale with a customer membership and redeem eligible rewards after validating the rules.

• Consent-based customer communications and appropriate controls for personal data.

The MVP should start with straightforward loyalty rules and store-specific offers. Advanced segmentation, automated campaigns and third-party marketing integrations can follow after launch.

6. Multi-Store Analytics

• Daily, weekly and monthly sales; transaction count and average bill value.

• Gross profit and margin based on recorded cost of goods sold, visible only to authorized roles.

• Stock on hand, stock valuation, low-stock products and near-expiry inventory.

• Best-selling products, category performance and discount reporting.

• Branch-by-branch comparison and consolidated owner reporting.

• Purchases, supplier balances and expenses where included in the final accounting scope.

7. SaaS Subscription & Super Admin

• Retailer onboarding, business profile, branch creation and account lifecycle management.

• Monthly/annual subscription plans, plan status, trial periods if offered, and renewal/payment status.

• Configurable limits for branches, users or other plan features.

• Platform-level reporting for active businesses, subscriptions and product usage.

• Support tools and controlled account suspension/reactivation.

8. Proposed Technology & Architecture

Layer	Proposed approach
Web admin & POS	Flutter Web, responsive layouts for desktop/tablet
Mobile apps	Flutter for Android and iOS
Backend	Go with Gin framework; REST API
Database	PostgreSQL with GORM and tenant-scoped data access
Authentication	Secure sessions/token lifecycle, role and permission checks
Files & hosting	S3-compatible object storage and cloud hosting such as AWS

Every record must be scoped to the correct business and, where applicable, branch. Sensitive fields such as purchase cost must not be returned to cashier APIs. Server-side authorization, audit logging, backups, and secure handling of customer data are core requirements.

9. Recommended Delivery Phases

Phase	Scope	Outcome
1 — Foundation	Tenant setup, authentication, roles, branches, products, suppliers and purchase/stock ledger	Secure business foundation
2 — POS	Cart, discounts, payment records, invoice/PDF, printing and WhatsApp sharing	Retailers can transact
3 — Reports & loyalty	Profit/stock analytics, customer shop codes, offers, points and redemption	Owner insights and customer retention
4 — Apps & launch	Customer/owner mobile experience, SaaS subscription workflows, QA and deployment	Launch-ready platform

Planning estimate only: a focused MVP may take approximately 8–12 weeks; a broader production release may take around 3–5 months. Actual delivery depends on team size, detailed acceptance criteria, integrations, offline needs, printing hardware and tax requirements.

10. Key Acceptance Criteria

• A business cannot access another business’s data, even by changing IDs in API requests.

• Cashier responses never expose purchase cost or profit fields.

• A cashier can create a sale and apply either a fixed ■ discount or a percentage discount within configured limits.

• Discounts, price edits, stock adjustments, refunds and cancellations are auditable.

• Selling a product updates the correct branch’s stock and records the cost basis for reporting.

• A customer can join a shop using its code/QR and see that shop’s valid offers and rewards.

• An owner can view authorized consolidated reports across their own branches.

• Invoices can be generated and shared, and supported printer configurations are tested before launch.

This document describes the proposed product scope. Final estimates, legal/tax compliance, privacy requirements, integrations and implementation details should be confirmed during discovery and technical design.